

Title

A triple-blind sham-controlled design for evaluating non-invasive brain stimulation in alcohol dependence

Abstract

Background: Limited pharmaceutical options exist for individuals with severe alcohol dependence, creating need for novel interventions. Non-invasive neuromodulation techniques may offer adjunctive benefit. This protocol describes a prospective, multi-site trial examining whether direct current stimulation can reduce drinking behaviors in adults with alcohol use disorder. **Methods:** We will recruit 350 adults with confirmed alcohol use disorder from 14 clinical sites. Participants will be randomly assigned 1:1 to receive either active stimulation (electrodes positioned at F4/F3, 2 mA current) delivered daily for five consecutive days, or matched sham stimulation. Both groups will undergo follow-up assessments every 4 weeks throughout the 24-week study period. Randomization will be stratified by site and baseline consumption severity. Participants, assessors, and clinicians will be blinded to treatment allocation. Primary outcomes will include drinking frequency metrics and weekly alcohol volume. **Eligibility:** adults aged 30-70 with documented alcohol use disorder, stable housing, capacity to provide informed consent. **Study design:** triple-blind, parallel-group, randomized controlled trial.